CLINICOS · ARCHITECTURE & DESIGN

The Patient File

Turning ClinicOS into the practice's system of record

PLAN A — depth / system of record

Document: High-level system design — full-system UML, architecture & use-case views

Companion spec: `ClinicOS/SystemOfRecordSpec.md` (detailed build spec)

Prepared: 7 August 2026 · **Status:** Proposed

01 How the system works with Plan A

Plan A does not replace ClinicOS — it extends it. The queue stays the front door; the new Patient File turns the data that door already collects into the record the practice works out of every day.

The shift in one sentence

Today ClinicOS is a walk-in **queue** that sits *beside* a practice's workflow. With Plan A it becomes the practice's **system of record** that sits *inside* the workflow — the surface every consultation runs on.

The loop, end to end

Patient joins queue → **Intake captured** → **Doctor opens Patient File** → **Writes consult note** → **Record grows**

Every visit already creates an appointment, intake answers and a timestamped event trail. Plan A **surfaces** that as a patient timeline, adds the **consult note** the doctor writes during the visit, and attaches the **digitised historical file** — so the record both starts full (from digitisation) and keeps growing for free (from queue usage).

> **Why the note is the pivot.** The consult note is what pulls the daily workflow inside ClinicOS. Without it, the practice merely *looks things up* occasionally (Plan B — a cancellable filing cabinet). With it, the practice cannot run a consultation without ClinicOS — that is the retention.

What is new vs what already exists

ALREADY IN THE SYSTEM	ADDED BY PLAN A
Clinics, staff & roles · patients · appointments · intake responses · event audit log · visit feedback · the whole queue flow	Clinical summary on the patient · **consult notes** · **document store** (digitised files) · **patient consent** · **record-access auditing**

> **Compliance is part of the architecture, not a bolt-on.** Holding active clinical data puts ClinicOS under POPIA (special personal information) and HPCSA record-keeping. Consent capture and access auditing are therefore first-class components below — and they are also the moat that makes ClinicOS hard to clone.

02 System architecture

Clients talk to Supabase two ways: the patient app through token-scoped Edge Functions (no login), staff through the authenticated portal under row-level security. Plan A adds a private document store, two document Edge Functions, and the digitisation ingest path.

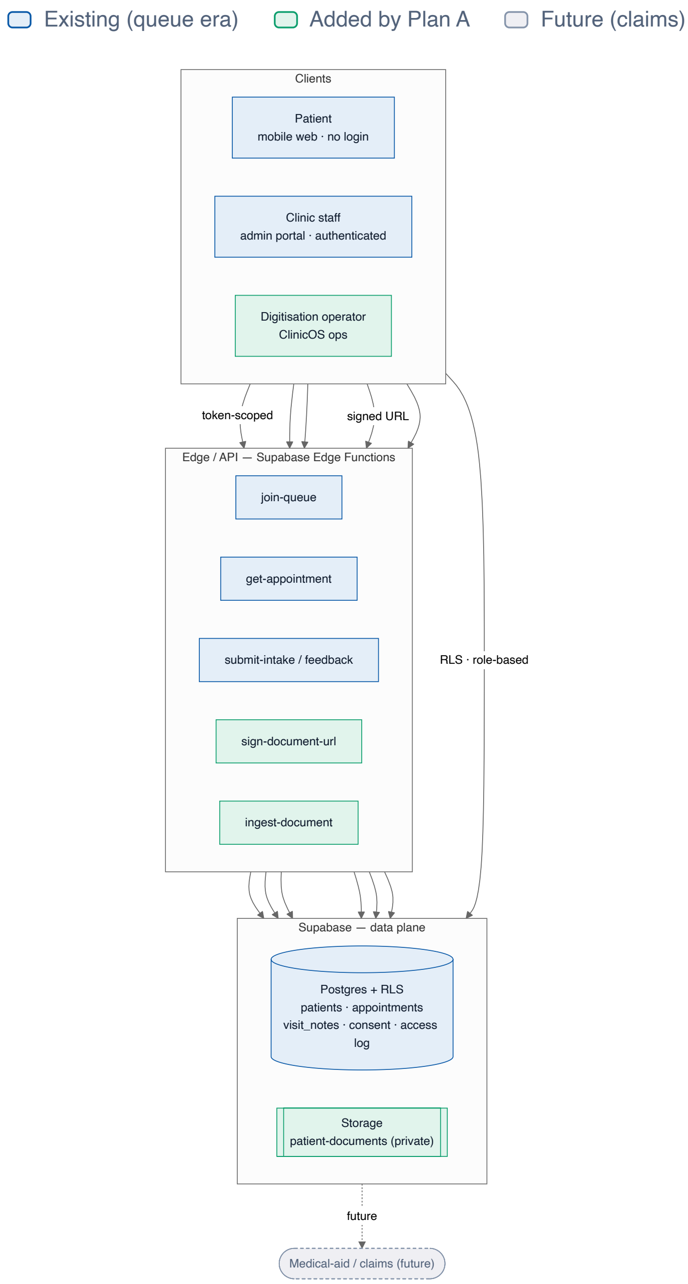

Figure 1 — Deployment / component architecture. Green components are introduced by Plan A.

03 Full-system UML (domain model)

The complete data model with Plan A in place. The queue-era entities are unchanged; Plan A adds four entities and a clinical summary on the patient. Everything remains scoped to a clinic for multi-tenant isolation.

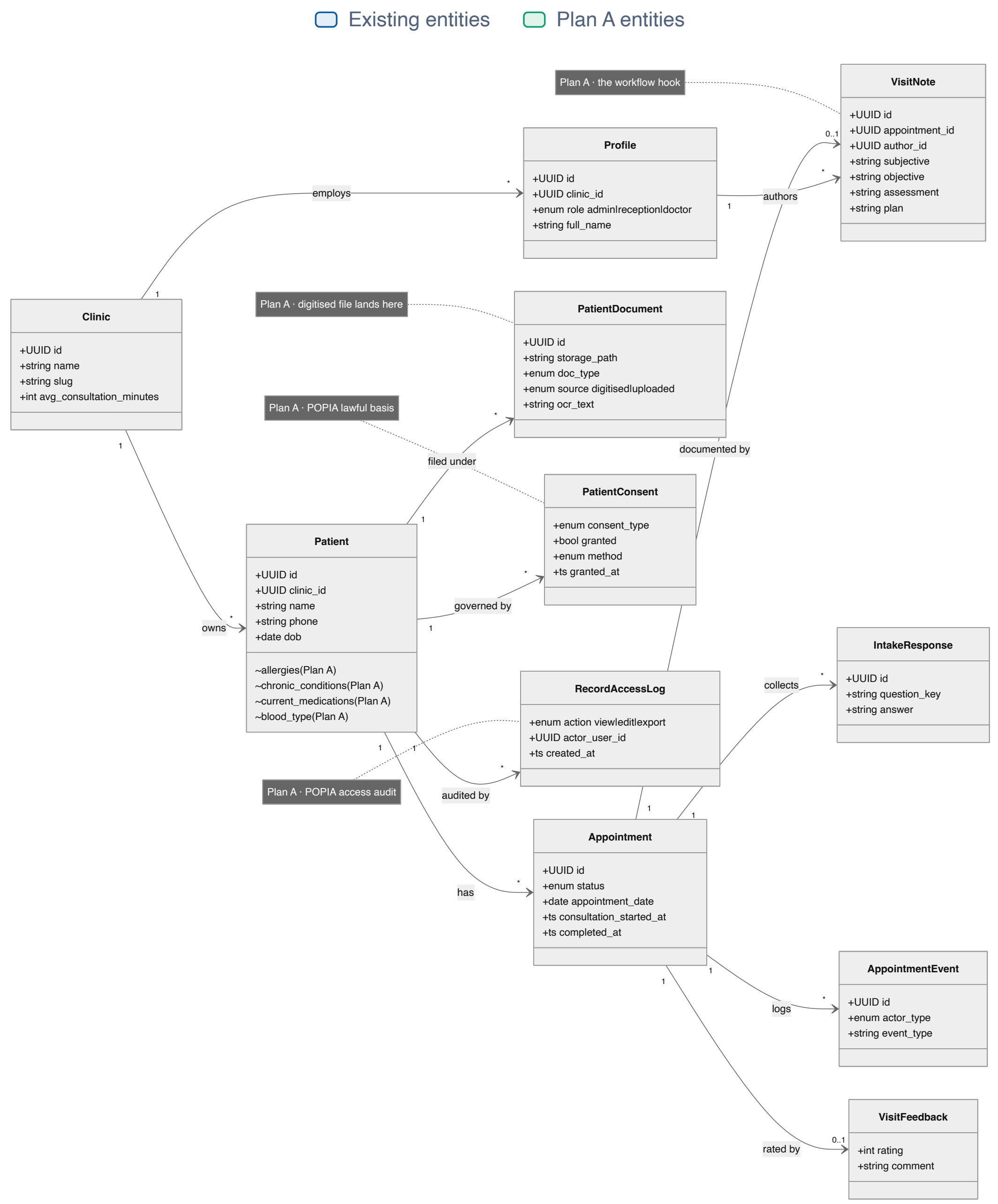

Figure 2 — Domain model. Plan A entities: VisitNote, PatientDocument, PatientConsent, RecordAccessLog, plus clinical fields (~) on Patient.

04 Use-case view

Who does what within ClinicOS once Plan A ships. Patient-facing actions stay friction-free; the new clinical and compliance use cases belong to staff and ClinicOS operations.

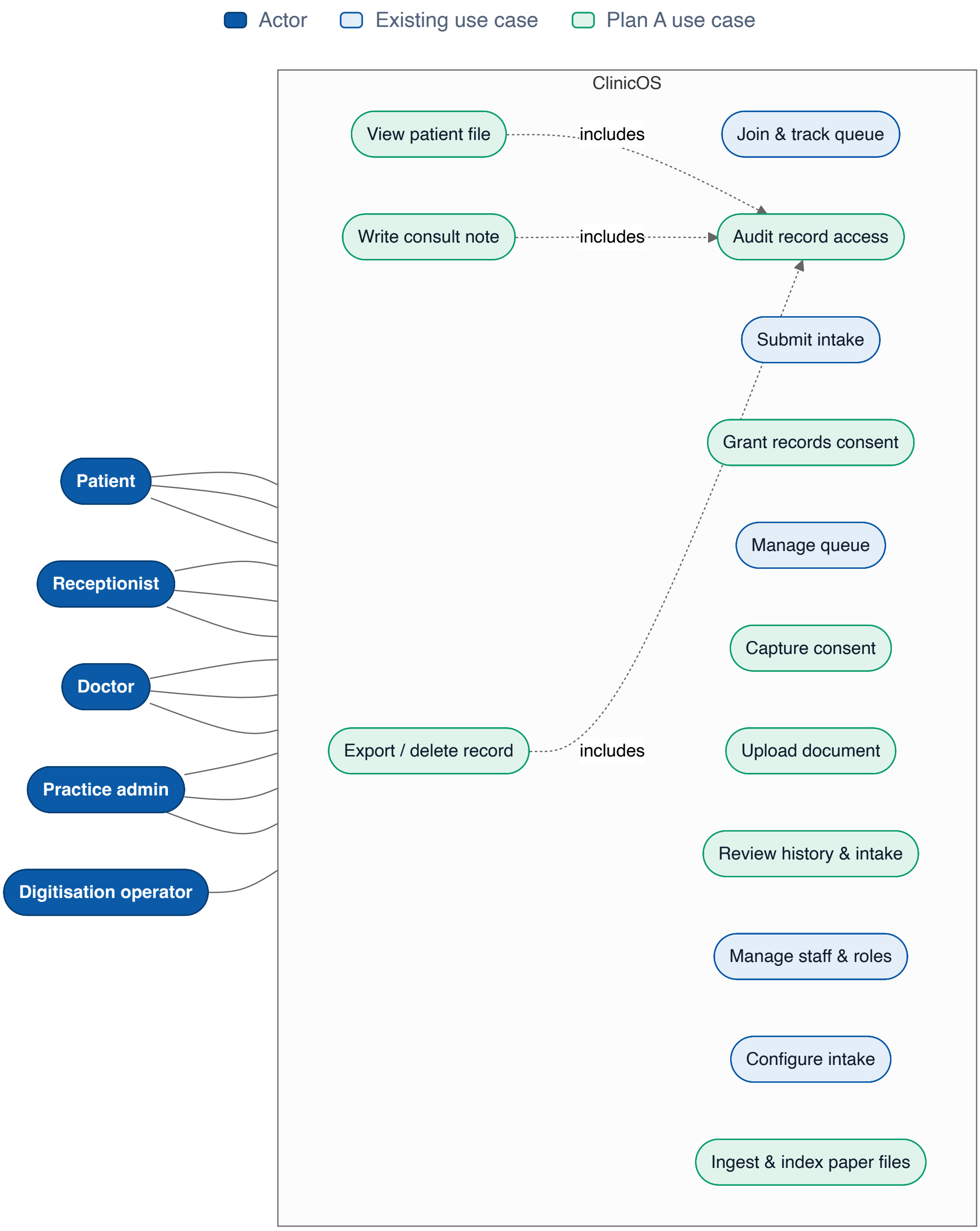

Figure 3 — Use-case view. Green use cases are introduced by Plan A; access auditing (u14) is *included* by every record interaction for POPIA.